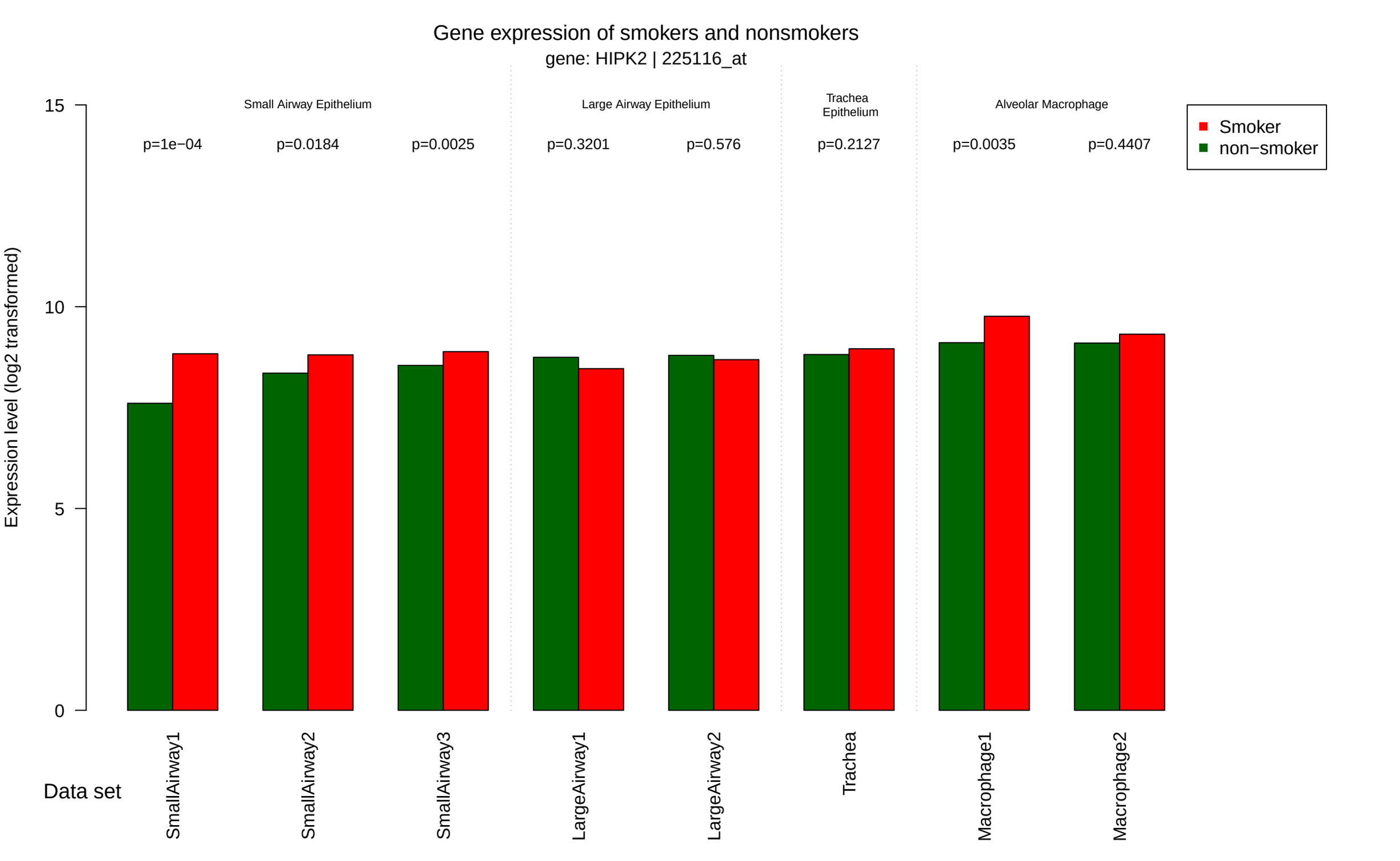
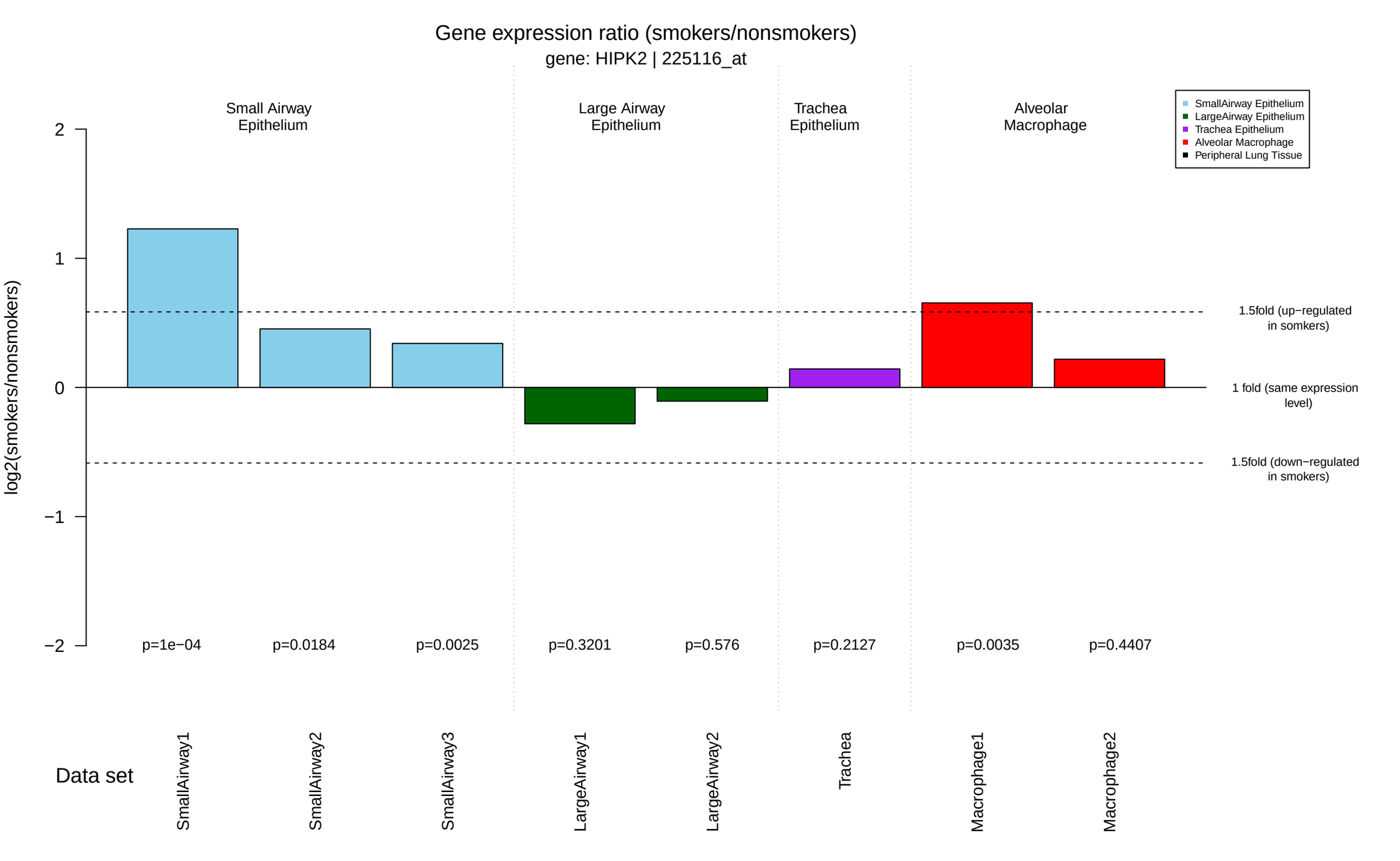
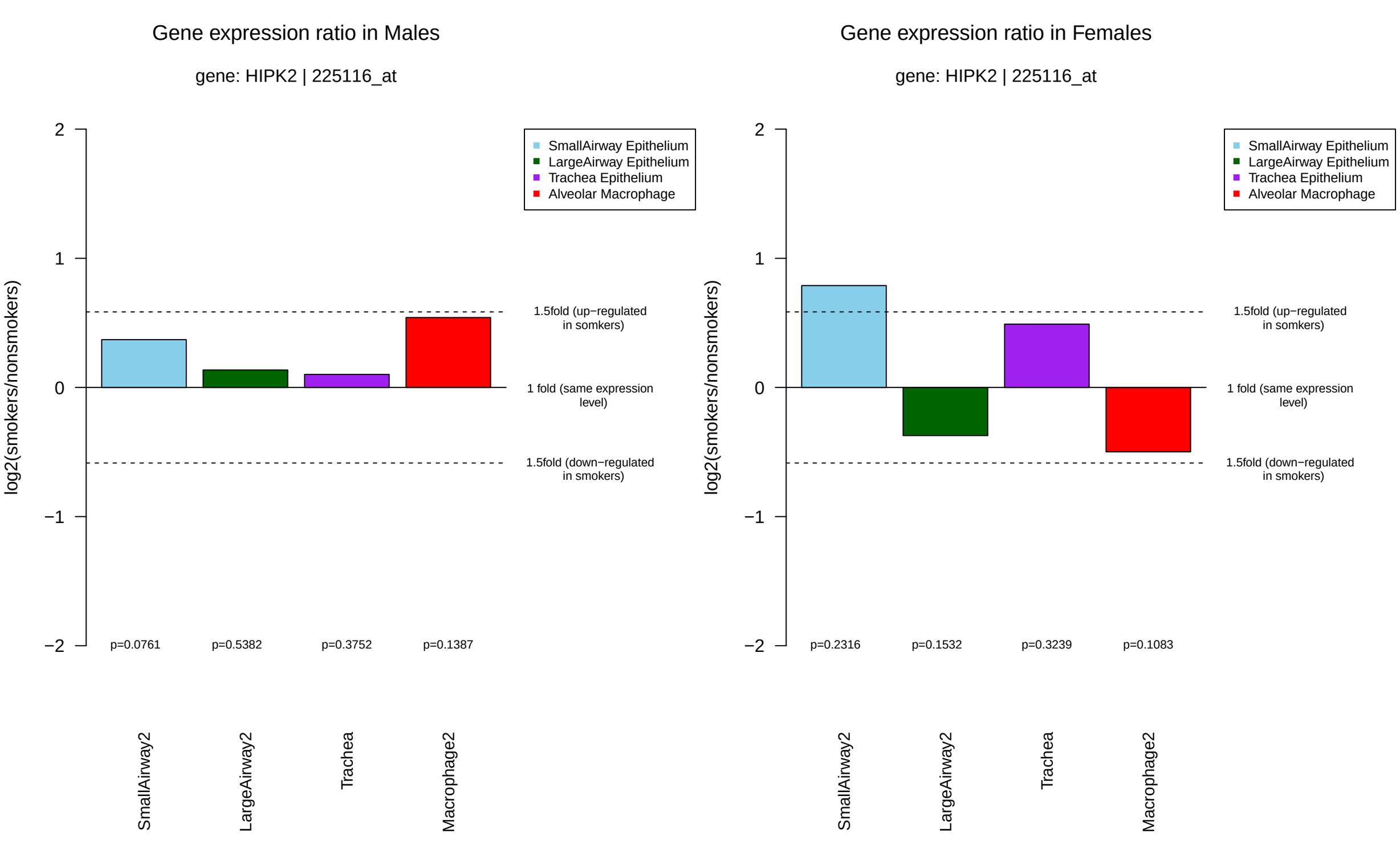
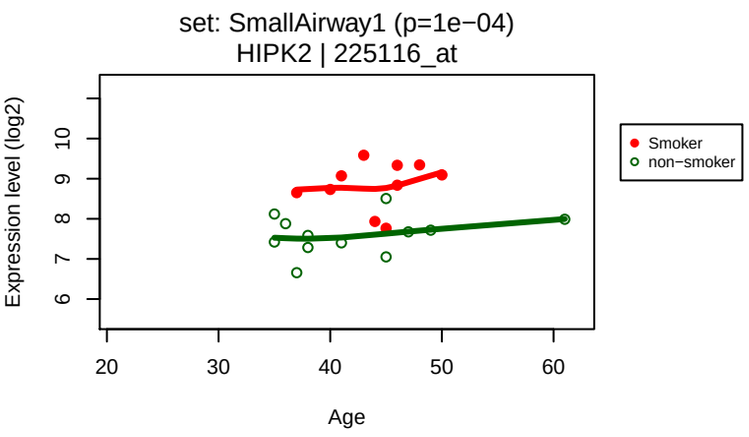
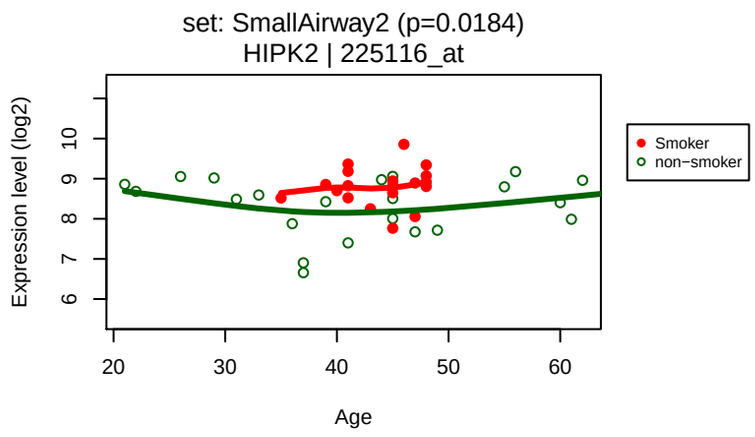
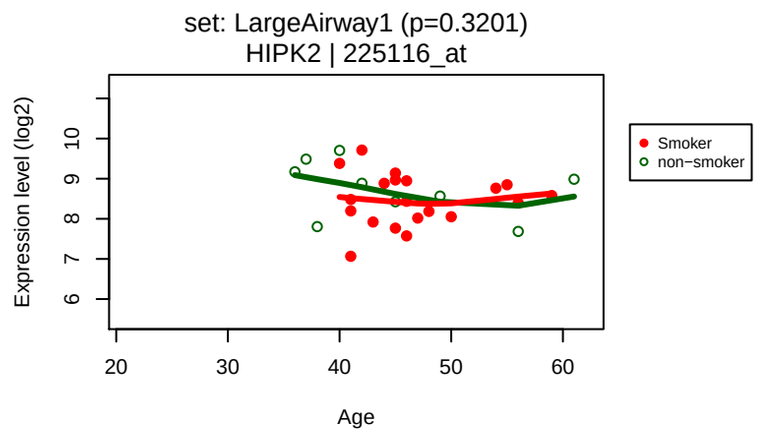

Gene expression of smokers and nonsmokers
gene: HIPK2 | 225116_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=1e−04
p=0.0184
p=0.0025
p=0.3201
p=0.576
p=0.2127
p=0.0035
p=0.4407
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

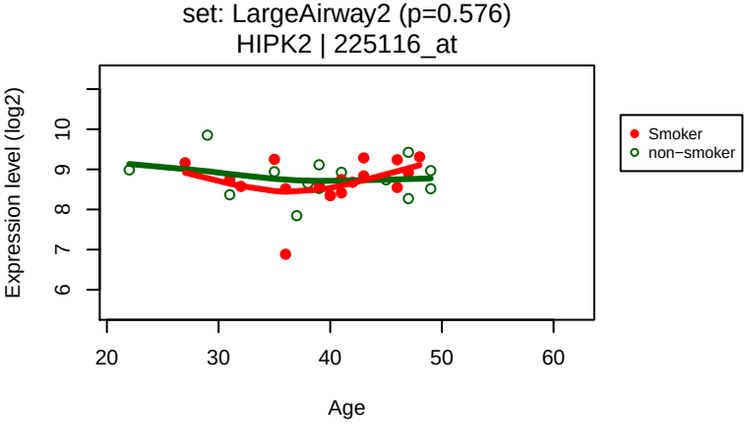

Gene expression ratio (smokers/nonsmokers)
gene: HIPK2 | 225116_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.0184
p=0.0025
p=0.3201
p=0.576
p=0.2127
p=0.0035
p=0.4407
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

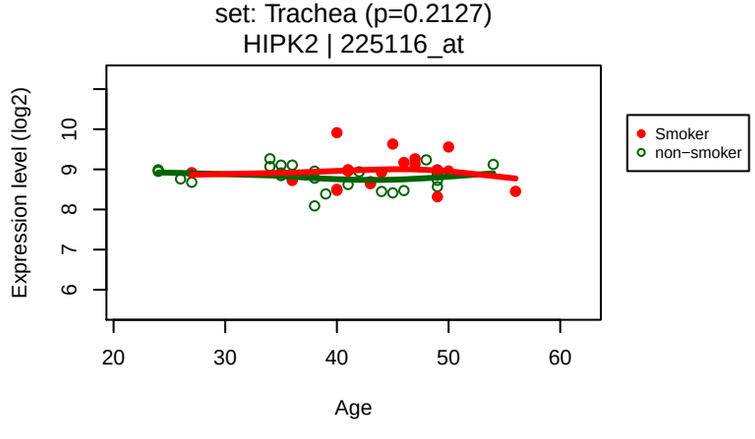

Gene expression ratio in Males
gene: HIPK2 | 225116_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0761
p=0.5382
p=0.3752
p=0.1387
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIPK2 | 225116_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2316
p=0.1532
p=0.3239
p=0.1083
SmallAirway2
LargeAirway2
Trachea
Macrophage2

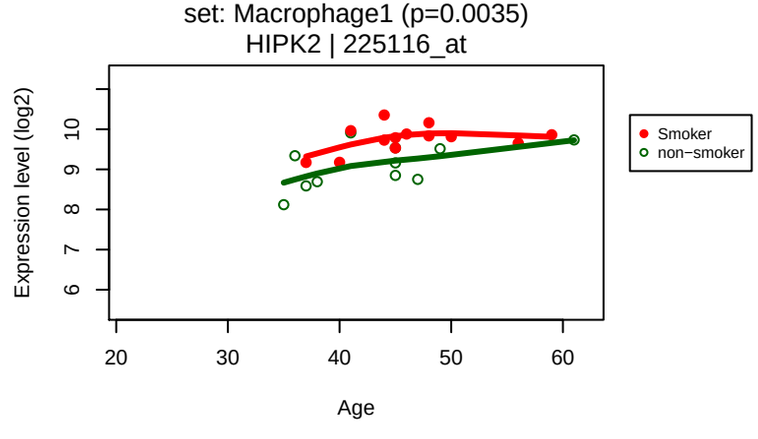

set: SmallAirway1 (p=1e−04)
HIPK2 | 225116_at
Expression level (log2)
Age
Smoker
non−smoker

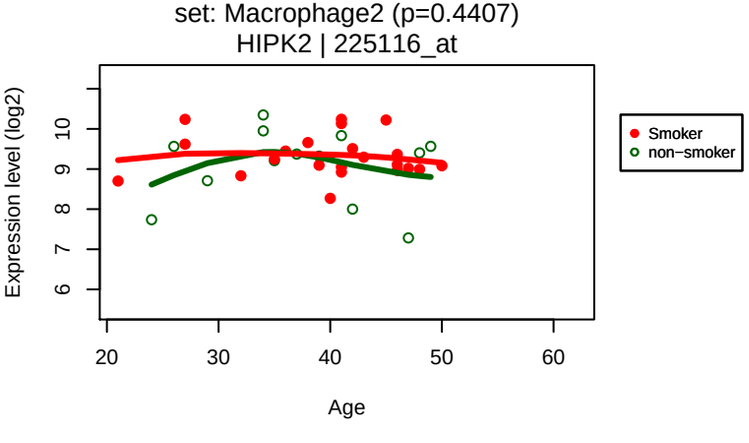

set: SmallAirway2 (p=0.0184)
HIPK2 | 225116_at
Expression level (log2)
Age
Smoker
non−smoker

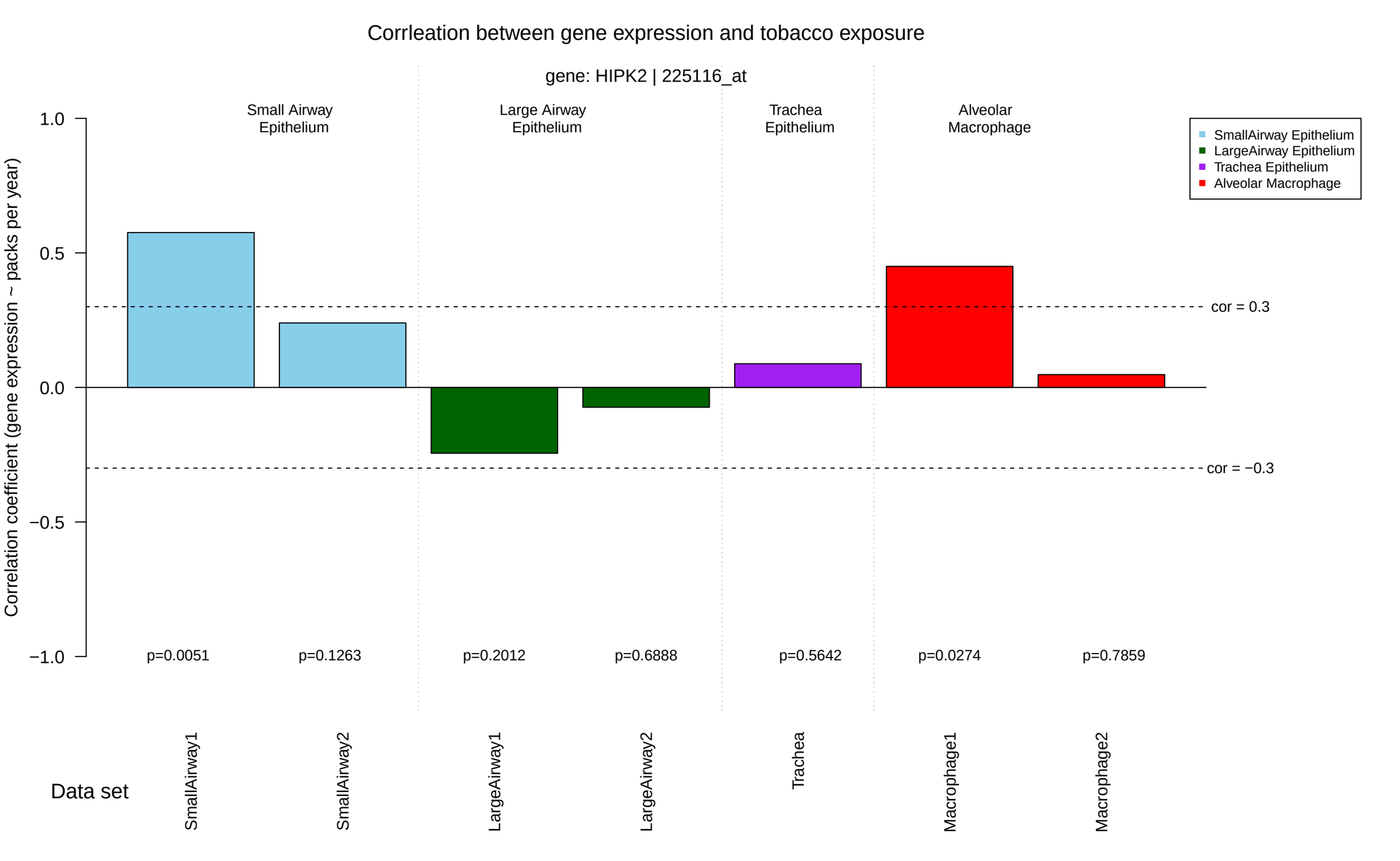

set: LargeAirway1 (p=0.3201)
HIPK2 | 225116_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.576)
HIPK2 | 225116_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2127)
HIPK2 | 225116_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0035)
HIPK2 | 225116_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4407)
HIPK2 | 225116_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIPK2 | 225116_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0051
p=0.1263
p=0.2012
p=0.6888
p=0.5642
p=0.0274
p=0.7859
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2